

Week 2 portfolio

Inserting:

Insert Document

To collection lab2.PEOPLE

```
1  _id: ObjectId('6ac4f49bc227abf86fc4986c')
2  firstName: "bob //"
3  lastName: "smith //"
4  gender: "Male //"
5  age: 29
6  email: "bob@gmail.com //"
7  education: "Bachelor //"
8  salary: 9000
9  martialStatus: "Single //"
```

VIEW



ObjectId
String
String
String
Int32
String
String
Int32
String

Updating:

```
1  _id: ObjectId('6ac4f010c227abf86fc497ae')
2  firstName: "Rebecca //"
3  lastName: "Douglas //"
4  gender: "Female //"
5  age: 20
6  email: "r.douglas@randatmail.com //"
7  education: "Bachelor //"
8  salary: 8283
9  maritalstatus: "Single //"
```

Document modified.

CANCEL UPDATE

Aggregation task 1:

Stage1 \$match

```
1  /**
2  * query: The query in MQL.
3  */
4  {
5    education:
6    "Bachelor",
7    age: {$gte:21}
8  }
```

Output preview after \$match stage (Sample of 10 documents)

```
_id: ObjectId('6ac4f010c227abf86fc497a4')
firstName: "Grace"
lastName: "Nelson"
gender: "Female"
age: 21
email: "g.nelson@randatmail.com"
education: "Bachelor"
salary: 5347
maritalstatus: "Single"
```

```
_id: ObjectId('6ac4f010c227abf86fc497aa')
firstName: "Charlie"
lastName: "Perkins"
gender: "Male"
age: 28
email: "c.perkins@randatmail.com"
education: "Bachelor"
salary: 3586
maritalstatus: "Single"
```

```
_id:
firs
last
gend
age:
educ
sala
mari
```

Aggregation task 2:

Week 2 portfolio

Stage 2

```
1 /**
2  * _id: The id of the group.
3  * fieldN: The first field name.
4  */
5 {
6   _id: "$gender",
7   Avg: { $avg: "$age" },
8   minAge: { $min: "$age" },
9   maxAge: { $max: "$age" }
10 }
```

Output preview after [\\$group](#) stage (Sample of 2 documents)

```
_id: "Male"
Avg : 25.923076923076923
```

```
_id: "Female"
Avg : 25
```

Stage 2

```
1 /**
2  * _id: The id of the group.
3  * fieldN: The first field name.
4  */
5 {
6   _id: "$gender",
7   Avg: { $avg: "$age" },
8   minAge: { $min: "$age" },
9   maxAge: { $max: "$age" },
10 }
```

Output preview after [\\$group](#) stage (Sample of 2 documents)

```
_id: "Male"
Avg : 25.923076923076923
minAge : 22
maxAge : 30
```

```
_id: "Female"
Avg : 25
minAge : 21
maxAge : 29
```

Stage 2

```
1 /**
2  * _id: The id of the group.
3  * fieldN: The first field name.
4  */
5 {
6   _id: "$gender",
7   Avg: { $avg: "$age" },
8   minAge: { $min: "$age" },
9   maxAge: { $max: "$age" },
10   MaxSalary: { $max: "$salary" },
11   MinSalary: { $min: "$salary" },
12   AvgSalary: { $avg: "$salary" }
13 }
```

Output preview after [\\$group](#) stage (Sample of 2 documents)

```
_id: "Male"
Avg : 25.923076923076923
minAge : 22
maxAge : 30
MaxSalary : 9759
MinSalary : 1260
AvgSalary : 5540.692307692308
```

```
_id: "Female"
Avg : 25
minAge : 21
maxAge : 29
MaxSalary : 8799
MinSalary : 509
AvgSalary : 5020.846153846154
```

Lab task 1

Stage 2

```
1 /**
2  * query: The query in MQL.
3  */
4 {
5   education: "Master"
6 }
7 }
```

Output preview after [\\$match](#) stage (Sample of 10 documents)

```
_id: ObjectId('6ac4f010c227abf86fc497b3')
firstName: "Evelyn"
lastName: "Wells"
gender: "Female"
age: 24
email: "e.wells@randatmail.com"
education: "Master"
salary: 2923
maritalstatus: "Single"
```

```
_id: ObjectId('6ac4f010c227abf86fc497b4')
firstName: "Martin"
lastName: "Alexander"
gender: "Male"
age: 26
email: "m.alexander@randatmail.com"
education: "Master"
salary: 2739
maritalstatus: "Single"
```

Stage 2

```
1 /**
2  * _id: The id of the group.
3  * fieldN: The first field name.
4  */
5 {
6   _id: "$maritalstatus",
7   Avg: { $avg: "$age" },
8   minAge: { $min: "$age" },
9   maxAge: { $max: "$age" },
10   MaxSalary: { $max: "$salary" },
11   MinSalary: { $min: "$salary" },
12   AvgSalary: { $avg: "$salary" }
13 }
```

Output preview after [\\$group](#) stage (Sample of 2 documents)

```
_id: "Married"
Avg : 25.8
minAge : 18
maxAge : 30
MaxSalary : 8483
MinSalary : 940
AvgSalary : 5227.5
```

```
_id: "Single"
Avg : 25.333333333333332
minAge : 18
maxAge : 30
MaxSalary : 8722
MinSalary : 718
AvgSalary : 3783.8
```

Lab task 2:

Week 2 portfolio

Stage 1 **\$match**

```
1 /**
2  * query: The query in MQL.
3  */
4
5 {
6   gender: "Female"
7 }
8
```

Output preview after **\$match** stage (Sample of 10 documents)

<pre>{ "_id": ObjectId('6ac4f010c227abf86fc497a4'), "firstName": "Grace", "lastName": "Nelson", "gender": "Female", "age": 21, "email": "g.nelson@randatmail.com", "education": "Bachelor", "salary": 5347, "maritalstatus": "Single" }</pre>	<pre>{ "_id": ObjectId('6ac4f010c227abf86fc497a7'), "firstName": "Tiana", "lastName": "Fowler", "gender": "Female", "age": 27, "email": "t.fowler@randatmail.com", "education": "Primary", "salary": 3529, "maritalstatus": "Married" }</pre>	<pre>{ "_id": ObjectId('6ac4f010c227abf86fc497a8'), "firstName": "Daryl", "lastName": "Johnson", "gender": "Male", "age": 20, "email": "d.johnson@randatmail.com", "education": "Upper secondary", "salary": 4450, "maritalstatus": "Married" }</pre>
---	---	---

Stage 2 **\$group**

```
1 /**
2  * _id: The id of the group.
3  * fieldN: The first field name.
4  */
5 {
6   _id: "$gender",
7   Avg: { $avg: "$age" },
8   minAge: { $min: "$age" },
9   maxAge: { $max: "$age" },
10  MaxSalary: { $max: "$salary" },
11  MinSalary: { $min: "$salary" },
12  AvgSalary: { $avg: "$salary" }
13 }
```

Output preview after **\$group** stage (Sample of 1 document)

```
{
  "_id": "Female",
  "Avg": 24.064814814814813,
  "minAge": 18,
  "maxAge": 30,
  "MaxSalary": 9925,
  "MinSalary": 509,
  "AvgSalary": 4954.861111111111
}
```

Lab task 3:

Stage 1 **\$match**

```
1 /**
2  * query: The query in MQL.
3  */
4
5 {
6   gender: "Male"
7 }
8
```

Output preview after **\$match** stage (Sample of 10 documents)

<pre>{ "_id": ObjectId('6ac4f010c227abf86fc497a5'), "firstName": "Justin", "lastName": "West", "gender": "Male", "age": 27, "email": "j.west@randatmail.com", "education": "Doctoral", "salary": 5783, "maritalstatus": "Married" }</pre>	<pre>{ "_id": ObjectId('6ac4f010c227abf86fc497a6'), "firstName": "Daryl", "lastName": "Johnson", "gender": "Male", "age": 20, "email": "d.johnson@randatmail.com", "education": "Upper secondary", "salary": 4450, "maritalstatus": "Married" }</pre>
---	---

Stage 2 **\$group**

```
1 /**
2  * _id: The id of the group.
3  * fieldN: The first field name.
4  */
5 {
6   _id: "$gender",
7   Avg: { $avg: "$age" },
8   minAge: { $min: "$age" },
9   maxAge: { $max: "$age" },
10  MaxSalary: { $max: "$salary" },
11  MinSalary: { $min: "$salary" },
12  AvgSalary: { $avg: "$salary" }
13 }
```

Output preview after **\$group** stage (Sample of 1 document)

```
{
  "_id": "Male",
  "Avg": 24.440860215053764,
  "minAge": 18,
  "maxAge": 30,
  "MaxSalary": 9989,
  "MinSalary": 826,
  "AvgSalary": 5201.4731182795695
}
```

Lab task 4 :

Week 2 portfolio

Stage 2 

```
1  ▾ /**
2    * _id: The id of the group.
3    * fieldN: The first field name.
4    */
5  ▾ {
6    ▾ _id: {
7      gender: "$gender",
8      maritalStatus: "$maritalStatus"
9    },
10    count: { $sum: 1 }
11  }
12
```

Output preview after [\\$group](#) stage (Sample of 2 documents)

▾ _id: Object
gender: "Female"
count : 108

▾ _id: Object
gender: "Male"
count : 93

Reflection:

I learned quite a lot in today's lesson. We not only got to use mongodb but also added filters to help thin down the search for specified users which would really be useful in actual data companies.